The Infinite Loop Considered Harmless

The lambda calculus derived from itself, over one symbol

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The bent card. The accomplice marks one card so the mark believes the quarter can be followed; this is that card, and the throw unbends it.

Reading note

The object of this volume is the grammar with one terminal T and the calculus in which its sentences are terms. Three facts organize it. The sentences of the grammar, read with T as one application of a bound function, are the Church numerals. The term $\mathbf{\Omega} = (\lambda x.xx)(\lambda x.xx)$ has a reduction sequence whose trace, one terminal per step, is T^n for every n, and so carries exactly the structure of a Peano counter. A typed calculus in which every term terminates cannot write $\mathbf{\Omega}$ or a general fixed-point combinator; it still admits every recursion whose termination it can establish, and a search whose termination it cannot establish is closed by a datum adjoined from outside its terms. That datum is the *bent card* (Definition 3.7).

Section 1 fixes the calculus and its numerals and identifies the grammar with them. Section 2 reproduces the self-interpreter. Section 3 proves that $\mathbf{\Omega}$ is harmless under the normal-form acceptance policy (Definition 3.4) and determines exactly what reading it as a value costs. Section 4 runs a divergent process on purpose, as a search, and separates the search from the certificate it returns. Section 5 identifies the three bent cards the middle card requires and the one difference none of them absorbs, and Section 6 states what is proposed.

Definition 0.1 (Calibration)**.** A *calibration* of a statement is the stated source of its warrant, reported with the statement as part of its reading. This volume uses four.

Standard
The statement is cited from the literature with its hypotheses.

Proved here
The statement is proved in this volume; any premise cited from the literature or from a companion volume is named where it is used.

Measured
The statement is extracted from the declarations of the middle card, or observed in a run of it; the text says which. It is not a laboratory measurement unless a citation says so.

Proposed
The statement is an identification or continuation supplied by the reader and not derived. A proposed statement asserts that an isomorphism exists, between the numbering of the construction's descriptions by the natural numbers and the description of what it is identified with, that the machine can exploit as a computation. Both sides are finite records: the reading is a computable number, and the reference is a value with its uncertainty or a stated structure. The statement is therefore one the machine can evaluate, a case that could be false, and the comparison that would revise it is that evaluation. What no evaluation supplies is that the reference's finite record stands for what it records, as a finite halt stands for its limit.

Each label names the reference against which the statement's reading is calibrated.

The companion volumes are the geometric card, which fixes what must be retained, and the numerical card, which fixes what may be claimed from a reading; the middle card is a construction checked by a proof assistant. No declaration of the middle card appears here by name. In the game the bent card is the card with a corner turned up, so that the mark believes the quarter can be followed under it; the bend is a receipt supplied from outside, not a reading of the card. The throw that unbends it is Proposition 5.1: the reader that keeps the slot changes the walk's update from $(13, 18, 2)$ to $(14, 19, 3)$, and the walk converges to no point.

1 The calculus from one symbol

Definition 1.1 (Terms)**.** Terms are generated by three rules: a variable x is a term; if M is a term and x a variable then $\lambda x.M$ is a term (abstraction); if M and N are terms then MN is a term (application). The one computation rule is β: $(\lambda x.M)N \to M[x := N]$ [3, 1]. We write $\twoheadrightarrow$ for the reflexive–transitive closure of $\to$, $=_\beta$ for the generated equivalence, and $\mathrm{FV}(M)$ for the free variables of M. A term is in *normal form* if it contains no redex. *Standard.*

Definition 1.2 (Numerals)**.** For $n \geq 0$, $\overline{n} = \lambda f.\lambda x.\, f^{\,n}x$, where $f^{\,n}x$ is f applied n times to x. *Standard.*

Definition 1.3 (The one-terminal grammar)**.** $G = (\{S\}, \{\mathsf{T}\}, \{S \to \mathsf{T}S,\ S \to \epsilon\}, S)$. Its sentences are the words T^n, $n \geq 0$; its sentential forms are the words $\mathsf{T}^n S$ and T^n.

Proposition 1.4 (The one-terminal grammar is the numerals)**.** *Let the grammar of the companion card have terminal* T *and let* T *be read as application of* f*. Then every sentence of the grammar whose nonterminals are all erased is* $f^{\,n}x$ *for the number* n *of terminals it contains, and the reader that counts terminals is the map* $\overline{n} \mapsto n$ *obtained by applying* $\overline{n}$ *to the successor and zero.* Proved here.

Proof. Erasing nonterminals leaves a string of n copies of T; reading each as one application of f gives $f^{\,n}x$, and abstracting f and x gives $\overline{n}$. Applying $\overline{n}$ to succ and 0 gives $\mathrm{succ}^{\,n}0 = n$. □

The abstraction λf is the grammar's requirement of a name for its symbol; the application that supplies succ is a choice made outside the numeral. The numeral determines how often f is applied and nothing about what f is.

Proposition 1.5 (Two readings, two arithmetics)**.** *Addition and multiplication of numerals are*

$$\mathbf{A} = \lambda m.\lambda n.\lambda f.\lambda x.\, mf(nfx), \qquad \mathbf{M} = \lambda m.\lambda n.\lambda f.\, m(nf).$$

Two readers of T^n *are defined by folding over the word:*

(a) the tally*, which starts at* $\overline{0}$ *and applies* $v \mapsto \mathbf{A}\,\overline{1}\,v$ *once per terminal, reads* $\overline{n}$*;*

(b) the positional *reader, the binary Horner fold, which starts at* $\overline{0}$ *and applies* $v \mapsto \mathbf{A}\,\overline{1}\,(\mathbf{M}\,\overline{2}\,v)$ *once per terminal, reads* $\overline{2^n - 1}$*.*

The tally uses addition alone; the positional reader also uses multiplication. Standard *for the arithmetic terms;* proved here *for the values of the two folds. The tally is the reader of Proposition 1.4, and the positional reader is the register of the companion volumes.*

Proof. By induction on n. The tally adds one per terminal. For the positional reader the value at $n = 0$ is $0 = 2^0 - 1$, and $2(2^n - 1) + 1 = 2^{n+1} - 1$. □

2 The calculus from itself

A term can be encoded as a term, and a term can evaluate that encoding. The construction below is Mogensen's self-interpreter [8]. *Standard; reproduced here.*

Definition 2.1 (Encoding)**.** With a, b, c fresh,

$$\ulcorner x \urcorner = \lambda a.\lambda b.\lambda c.\, a\, x, \qquad \ulcorner MN \urcorner = \lambda a.\lambda b.\lambda c.\, b \ulcorner M \urcorner \ulcorner N \urcorner, \qquad \ulcorner \lambda x.M \urcorner = \lambda a.\lambda b.\lambda c.\, c\, (\lambda x. \ulcorner M \urcorner).$$

Definition 2.2 (Fixed-point combinator)**.** $\mathbf{Y} = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx))$. For every f, $\mathbf{Y}f =_\beta f(\mathbf{Y}f)$: both sides reduce to $f\big((\lambda x.f(xx))(\lambda x.f(xx))\big)$. *Standard.*

Theorem 2.3 (Self-interpretation)**.** *Let*

$$\mathbf{E} = \mathbf{Y}\,(\lambda e.\lambda m.\, m\,(\lambda x.x)\,(\lambda p.\lambda q.\,(e\,p)(e\,q))\,(\lambda p.\lambda v.\, e\,(p\,v))).$$

Then $\mathbf{E} \ulcorner M \urcorner =_\beta M$ *for every term* M*.* Standard *[8].*

Proof. Induction on M. By the fixed-point equation,

$$\mathbf{E} \ulcorner M \urcorner =_\beta \ulcorner M \urcorner\,(\lambda x.x)\,(\lambda p.\lambda q.\,(\mathbf{E}p)(\mathbf{E}q))\,(\lambda p.\lambda v.\,\mathbf{E}(pv)).$$

For a variable this is $(\lambda x.x)x =_\beta x$. For an application it is $(\mathbf{E}\ulcorner M \urcorner)(\mathbf{E}\ulcorner N \urcorner) =_\beta MN$ by the induction hypothesis. For an abstraction it is $\lambda v.\mathbf{E}\big(\ulcorner M \urcorner[x := v]\big)$; since a variable is encoded by itself under a, $\ulcorner M \urcorner[x := v] = \ulcorner M[x := v] \urcorner$, and the induction hypothesis gives $\lambda v.M[x := v]$, which is $\lambda x.M$ up to renaming of the bound variable. □

The evaluator is a term of the calculus, the encoding is a term of the calculus, and the equation $\mathbf{E} \ulcorner M \urcorner =_\beta M$ is an equation of the calculus; this is the sense in which the calculus is derived from itself. $\mathbf{E}$ is closed. Applied to the encoding of a numeral it returns the numeral with its abstraction λf intact: the operational interpretation of f is supplied from outside the interpreter, as in Proposition 1.4.

3 The loop

Definition 3.1 (The term that never stops)**.** $\mathbf{\Omega} = (\lambda x.xx)(\lambda x.xx)$. Its only redex is the whole term, $\mathbf{\Omega} \to \mathbf{\Omega}$, and $\mathbf{\Omega}$ has no normal form [1]. *Standard.*

The self-interpreter inherits it: $\mathbf{E} \ulcorner \mathbf{\Omega} \urcorner$ has no normal form, and $\mathbf{Y}$ is built from the same self-application.

Proposition 3.2 (The loop's record is the one-symbol tape)**.** *Read each reduction step as writing one terminal. Then the record of* $\mathbf{\Omega}$ *after* n *steps is the string* T^n*, the tally reads* n*, and the positional reading is* $2^n - 1$*. No step writes a different symbol, and no prefix of the record determines any later step differently from the first. Under the tally the set of records, with the empty record and the extension by one step, is the Peano structure* $(\mathbb{N}, 0, \mathrm{succ})$. Proved here.

Proof. $\mathbf{\Omega}$ has exactly one redex and contracting it returns $\mathbf{\Omega}$, so the reduction sequence is unique and every step is the same step. One terminal per step gives T^n after n steps, and Propositions 1.4 and 1.5 give the two readings. Extension by one step is $\mathsf{T}^n \mapsto \mathsf{T}^{n+1}$, which the tally carries to $n \mapsto n + 1$; it is injective and never returns to the empty record. □

The record grows by one unchanged symbol per step; its positional reading grows exponentially while the term read never changes. The identification of this counter with the de Sitter counter of the numerical card is the subject of the first remark of Section 6.

Proposition 3.3 (The loop writes no quarter)**.** $\mathrm{FV}(\mathbf{\Omega}) = \emptyset$*, and no prefix of the record of* $\mathbf{\Omega}$ *determines a value of any variable. In particular the loop supplies no* f *to any numeral and decides no comparison.* Proved here.

Proof. $\mathbf{\Omega}$ is closed, and its reduction sequence contains only $\mathbf{\Omega}$; a record consisting of one closed term repeated contains no substitution for any variable and no branch. □

Definition 3.4 (Acceptance policy; harmless)**.** An *acceptance policy* is a rule stating when a record system accepts a term as the completed value of a computation. The *normal-form policy* accepts a term only when the computation has reached it and it is in normal form. A computation is *harmless* under a policy if the policy accepts from it no completed value that the computation has not reached; in particular a divergent computation is harmless under a policy exactly when the policy accepts no completed value from it.

Proposition 3.5 (Harm is a loop read as a value)**.**

(a) Under the normal-form policy $\mathbf{\Omega}$ *is never accepted as a value, so it is harmless, and every term has at most one value.* Standard*, by the Church–Rosser theorem [1].*

(b) Let M *be in normal form with* $x \notin \mathrm{FV}(M)$*. The term* $(\lambda x.M)\mathbf{\Omega}$ *reduces to* M *in one step and also has the infinite reduction sequence* $(\lambda x.M)\mathbf{\Omega} \to (\lambda x.M)\mathbf{\Omega} \to \cdots$*. A reader that accepts a term reached on an infinite reduction as a value therefore assigns* $(\lambda x.M)\mathbf{\Omega}$ *the two distinct values* M *and* $(\lambda x.M)\mathbf{\Omega}$*, and values are no longer unique.* Proved here.

(c) The calculus does not determine a value for $\mathbf{\Omega}$*: for every closed term* M *the theory* $\lambda\beta + \{\mathbf{\Omega} = M\}$ *is consistent [7, 1], and identifying every unsolvable term is consistent as well [1]. If* M *and* N *are distinct closed* $\beta\eta$*-normal forms, the theory* $\lambda\beta + \{\mathbf{\Omega} = M,\ \mathbf{\Omega} = N\}$ *equates every pair of terms, by Böhm's theorem [1].* Standard.

Proof. (a) The first clause is the definition and Definition 3.1; the second is uniqueness of normal forms. (b) Contracting the head redex gives $M[x := \mathbf{\Omega}] = M$; contracting $\mathbf{\Omega}$ in argument position returns the same term. M contains no redex and $(\lambda x.M)\mathbf{\Omega}$ does, so the two values differ. (c) The two theories together give $M = N$. □

A value for the loop is therefore an explicit selection: any one is consistent with the calculus, none is forced by it, and two different selections identify everything. The loop considered as a record is a counter (Proposition 3.2); the harm lies in a policy that takes an unfinished computation for a finished one. The normal-form policy keeps the arm that says "not yet" apart from the arm that says "done": a computation that has not halted is absence of evidence, and reading it as evidence of absence is the one mistake.

Proposition 3.6 (No loop in a terminating calculus)**.**

(a) In a typed λ*-calculus in which every typable term has a normal form, neither* $\mathbf{\Omega}$ *nor* $\mathbf{Y}$ *is typable.*

(b) In a consistent type theory with an empty type $\mathbf{0}$*, no closed term has the type* $\prod_A (A \to A) \to A$ *of a general fixed-point operator.*

Standard *[1].*

Proof. (a) $\mathbf{\Omega}$ has no normal form, and neither has $\mathbf{Y}$, whose body reduces to

$$\lambda f.f\big((\lambda x.f(xx))(\lambda x.f(xx))\big)$$

and so to an unbounded tower of f. (b) Such a term, applied to $\mathbf{0}$ and to the identity on $\mathbf{0}$, inhabits $\mathbf{0}$. □

A dependent type theory with inductive types, of the kind in which the middle card is written, is in case (b) relative to its consistency: it admits no general fixed-point operator, and a definition by recursion is accepted with a termination argument for it. Such theories admit a great many recursive definitions, by structural and by well-founded recursion, and solve many fixed-point equations by terminating constructions; a base case supplied to one of them is then computed, not adjoined. In the untyped calculus $\mathbf{Y}f =_\beta f(\mathbf{Y}f)$ supplies a fixed point by a combinator for every f; in the typed one, a recursion whose termination the theory does not establish is closed only by a datum.

Definition 3.7 (The bent card)**.** Let a terminating typed calculus carry a recursive search whose termination its terms do not establish. A *bent card* is a datum adjoined to the calculus from outside its terms, a base case, a declared fixed point, or a declared equality, that closes the search. It is a receipt: it is checked where it is used, and it is not computed by the calculus that uses it.

The bent card is the quarter in its λ-calculus form: the fixed point that the untyped calculus writes as a term and the typed calculus, for such a search, receives as a datum.

4 The search outside the record

A search is a divergent process run on purpose: it may not halt, and under the normal-form policy it is harmless (Proposition 3.5), provided that only what it finds is read.

Proposition 4.1 (A complete search confirms and does not exhaust)**.** *A first-order sentence in Skolem normal form is unsatisfiable if and only if some finite set of ground instances of its matrix, over the terms built from its own function symbols, is propositionally unsatisfiable. So an enumeration of the ground instances halts whenever there is a refutation to find, and in general runs forever when there is none. A refutation, once found, is a certificate that is checked in time polynomial in its size.* Standard *[5].*

The space of ground instances is the whole space about which the sentence can say anything. A search through it is the loop with a stopping condition: it writes one instance per step and stops only when what it has written closes.

Proposition 4.2 (Search and receipt are separate)**.** *A search that halts can hand a checker a certificate, and the checker reads the certificate without repeating the search. The value of the fifth Busy Beaver function was settled this way: programs enumerated the machines and decided each one, and a proof assistant checked the certificates [10].* Standard. *The middle card is built the same way: an elaborator searches, writes what it found as literals, and the checker confirms each line against the one before.* Measured, *extracted from the card's construction.*

The search is therefore never in the space the checker reads: its results are, its process is not. This is Proposition 3.5(a) for a loop meant to stop.

Forced and free. The steps of the middle card carry universe levels, one per *seat*. A seat is *forced* if the checker accepts the step at exactly one level of that seat, the others fixed, and *free* if it accepts the step at every level of that seat; a free seat is read by nothing in the step. The census is as follows. In the thirteenth step through the thirtieth, exactly two seats are free in each. From the step that first receives the local model, free seats accumulate and are inherited: four, five, seven, eight, nine, nineteen, thirty-five, nineteen. Raising every free seat of a step at once, each to a different level, is accepted as well, so the free seats are free jointly. One hundred and forty-nine seats are free in all. *Measured*, observed in 4635 single-seat changes and the joint changes. The forced levels are themselves computed from the description: each forced seat is the universe in which one construction keeps its quantum, set once and carried, and reading the steps for it reproduces the hand census on every step it covers. *Measured*, extracted from the declarations.

Each step's space of admissible assignments is therefore one forced point times every assignment of its free seats, and since levels range over $\mathbb{N}$ that space is infinite. The elaborator visits only the forced point and selects the minimal assignment on the free seats. A complete search would have to range over every assignment; like the enumeration of Proposition 4.1, it could confirm any member and never exhaust them. The minimal member, every free seat at the lowest level, is the one the middle card writes down: the shortest description of a point in a space no search finishes.

The lift. A forced seat raised by one level is a *lift*. The typeset record of the middle card climbs its thirty-six pages by thirty-five lifts, each page carrying the register of the page below one level higher, and its press check lowers the register thirty-five times before it reads it.

Proposition 4.3 (A lift carries no datum)**.** *Let $\uparrow\colon X \to X^{\uparrow}$ raise a type one level and $\downarrow\colon X^{\uparrow} \to X$ lower it. Then $\downarrow\uparrow x = x$ and $\uparrow\downarrow y = y$, so the two types are equivalent and the lift neither adds nor removes a distinction. Thirty-five lowerings return the register of page one unchanged.*

Proof. Both composites are the identity by the definition of the raised type, whose one field is the value raised; the thirty-five-fold composite is the identity by induction. □

Proved here; the count of thirty-five is *measured*, from the declarations.

So what a lift carries is not in the lift. It is a pointer to the step taken between two pages, and the mathematics of that step is performed at the pointer, on another card. The geometric card performs lifts 1 to 20, from the fact to the coupling of light to the electron; the numerical card performs lifts 21 to 35, from the coupling through the stationary system of light and matter to the laboratory and the press. Four of the thirty-five are bent cards in the sense of Definition 3.7, adjoined and not computed: the rule that takes the coefficient tape from the arm that cannot click (lift 11), the reading taken for the coupling (lift 20), the background, spinors, kinetic terms and mass (lift 26), and quantization with renormalization (lift 32). The other thirty-one are steps the cards perform. The search that found them is not in the record; the lifts that point to them are.

The flat region. For $a_0, a_1, \dots, a_n > 0$,

$$\int_0^{\infty} \prod_{k=0}^{n} \operatorname{sinc}(a_k x)\, dx = \frac{\pi}{2a_0} \qquad \text{whenever} \qquad a_1 + \cdots + a_n \le a_0,$$

and the integral is strictly smaller when $a_1 + \cdots + a_n > a_0$ [2]. *Standard.* With $a_k = 1/(2k+1)$ the condition is $1/3 + 1/5 + \cdots + 1/(2n+1) \le 1$. The partial sums are 0.955 through the factor $1/13$

and 1.022 through 1/15, so the integral equals $\pi/2$ through the factor 1/13 and falls below it at 1/15. *Proved here* for the sums. The proof of the cited result runs through convolution: each sinc transforms to a box, the product to a convolution of boxes, and the integral is that convolution's value at the origin, constant while the narrower widths fit inside the first. Repeated convolution of boxes builds splines; four boxes convolved give the cubic B-spline, piecewise cubic, so its fourth derivative vanishes between knots. Integration by parts, which is Stokes' theorem on a line, moves a derivative onto the test function and gives the weak form that spline elements solve; the natural cubic spline through given points is the curve of least bending energy [6], whose Euler–Lagrange equation is the beam equation, with fourth derivative zero where unloaded. *Standard*, each step. Four is also the count at which the companion cards read once around.

5 Three bent cards

The middle card, run to its end, requires three bent cards in the sense of Definition 3.7, and each is the step its search cannot take for itself.

The base case of the search. A carrier holds a slot it never reads. Let a tape t be folded onto a slot q, so that the record is the word $t \cdot q$, and define

$$L_{\mathrm{exact}}(t \cdot q) = |t \cdot q| - |q| = |t| \,.$$

This reader returns the tape's length whatever the slot holds. *Proved here*: the length of a concatenation is the sum of the lengths. Given that receipt, a walk read through a carrier of carriers is the walk already stored for the inner card, and the check is a lookup, finished in under a second. *Measured*, observed in a run.

Without the receipt, a reader that keeps the slot reads every length one cell longer.

Proposition 5.1 (The throw)**.** *Keeping the slot changes the coefficients of the walk's quadratic step from* $(13, 18, 2)$*, for which the step*

$$N(z) = \frac{13z^2 - 18}{2(13z - 18)}$$

is Newton's update for $13z^2 - 36z + 18$*, to* $(14, 19, 3)$*, for which*

$$G(z) = \frac{14z^2 - 19}{3(14z - 19)}$$

is no longer Newton's update for any quadratic. On the half-line $(19/14, \infty)$*, which* G *maps into itself, the only fixed point of* G *is the repelling point* $z_+ = (57 + \sqrt{1121})/56$*, and an orbit of* G *seeded at a rational point of the half-line converges to no point and does not tend to infinity.* Proved here.

Proof. The third coefficient no longer makes the step divide by the derivative. The fixed points of G solve $28z^2 - 57z + 19 = 0$ and are

$$z_\pm = \frac{57 \pm \sqrt{1121}}{56}, \qquad z_+ \approx 1.6157, \quad z_- \approx 0.4200,$$

with multipliers

$$G'(z) = \frac{14(14z^2 - 38z + 19)}{3(14z - 19)^2}, \qquad G'(z_+) \approx -2.0827, \quad G'(z_-) \approx 0.1494.$$

The upper fixed point, the counterpart of the reading, repels; the lower attracts, but it lies below the pole $19/14$ and is not reached from above it. On $(19/14, \infty)$ the only critical point of G is $z_c = (38 + \sqrt{380})/28 \approx 2.0533$, a minimum with $G(z_c) \approx 1.3689 > 19/14 \approx 1.3571$; since $G(z) \to \infty$ at both ends of the half-line, the half-line is invariant, and its only fixed point is z_+. An orbit seeded at a rational point above the pole therefore converges to no point: a limit would be a fixed point in the half-line, hence z_+, which repels, and a rational orbit never lands on the irrational z_+; nor does it tend to infinity, since $G(z)/z \to 1/3$. □

A double-precision computation outside the middle card shows the orbit from $z = 2$ wandering in [1.3689, 14.66] with Lyapunov exponent about 0.567. In exact rational arithmetic the unnormalized pair doubles its length every turn under either reader; in the reported runs only the reader that deducts the slot halted. Two runs of the reader that keeps the slot produced no reading and were stopped, one at 92 GB and one at 18 GB; in neither was the halting test met before the stop, and the walk followed on ratios left the jar, the bracket the middle card computes from its own lengths (*Sacred Texts*, appendix on the sealed predictions). *Measured*, observed in runs: the non-convergence is Proposition 5.1, and that the stopping condition was not met is evidence of these runs, not a consequence of it. Under the policy that accepts a reading only when the halting test is met (Definition 3.4), the walk without the receipt returns no accepted reading and is harmless; with the receipt handed in it halts.

Definition 5.2 (Declared zero)**.** Let a process produce corrections c_k, and let P_τ be its stopping predicate at resolution τ. A correction is *declared zero at resolution* τ when it satisfies P_τ; the process then reports its current state. A declared zero does not assert that the exact correction vanishes.

The halting test. The walk's stopping predicate is that its correction falls below the last printed place, $\tau = 10^{-24}$; from there it holds the same pair. By the exact error law of Newton's step for this quadratic (the numerical card, its proposition on exact error laws), an error $e > 0$ on the upper side is followed by

$$e' = \frac{e^2}{2e + d}, \qquad d = \frac{6\sqrt{10}}{13},$$

which is positive. The remaining error at every finite step is therefore nonzero, and the test declares it zero at resolution 10^{-24}. *Proved here*, from the error law. The limit the test stands for is the root $\alpha = (18 + 3\sqrt{10})/13$. It is finitely representable, as this expression with exact arithmetic in $\mathbb{Q}(\sqrt{10})$; it has no terminating decimal, and no element of the walk's rational record equals it. The reals of the middle card's language are built as classes of sequences whose operations are not computable: naming the root as such a real, rather than by the expression above, spends the choice the card's reading never spends. *Standard* for the construction of the reals.

At the reporting boundary two predicates are in play: P_τ, "the correction is below 10^{-24}", which holds at the reported step, and P_0, "the remaining error is zero", which holds at no finite step. Reporting the state as the root reads the first as the second: a statement true at resolution τ is taken for one false exactly. That is the true-equals-false of the argument, and the second bent card. *Proposed*, that the halting test is the declared fixed point.

The calibration. The reading is a ratio of counts, $x_\alpha = 972(6 + \sqrt{10})/65 = 137.011\,290\,5\ldots$. To give it units, let the second be fixed by the caesium hyperfine frequency, and let one symbol

of length be the distance light travels in one period of that transition; the speed of light is then one symbol per period. Its value in metres per second was declared exact in 1983, and the Planck constant and the elementary charge in 2019 [9]. *Standard.* With those declarations handed in, the vacuum permeability $\mu_0 = 2\alpha h/(e^2 c)$ and the impedance of free space $Z_0 = \mu_0 c$ are proportional to the fine-structure constant and are the electromagnetic constants still measured. The reading puts the impedance at $376.798\,\Omega$ against the recommended $376.730\,\Omega$, while $\mu_0\varepsilon_0 = 1/c^2$ does not move. *Proved here* as arithmetic. The declared exact values are the third bent card.

Proposition 5.3 (What no bent card absorbs)**.** *With all three receipts handed in, the reading is the same under every reader that measures length, every slot, and every assignment of the free seats; under the declared constants it appears entirely in the impedance of free space; and there it differs from measurement by* 1.80 *parts in ten thousand, which measurement excludes at about one part in* 10^{10}*.*

Proved here for the first clause from L_{exact} and the invariances of the companion cards; *measured* for the free seats, observed in a run in which every free seat of steps fourteen through thirty-eight was moved off its value at once with the steps still checking alike; *proved here* for the impedance; *standard* for the exclusion. The comparison itself is stated in the numerical card, its comparison section.

The middle card is a finite, checked description, and every step it cannot take for itself is a receipt it names: the base case of its search, the zero its halting test declares, and the constants that give it units. Each is a fixed point handed in from outside. The residue of Proposition 5.3 is the one difference none of them absorbs: no convention, no declaration of exactness, and no choice of reader moves it. That the residue is physical, and not the description's own, is the knower's.

6 What is proposed

Remark (The counter is the loop)*.* The companion cards carry a counter that increases by exactly one per written entry, a register that is the positional reading of a one-symbol tape, and a verdict decided once at the top. Propositions 3.2 and 3.3 identify the counter with the record of $\mathbf{\Omega}$ read one terminal per step, and the register with its positional reading. The further identification of the loop's step index n with the global time of the de Sitter slicing, and of $2^n - 1$ with the scale factor, is *proposed*, as in the numerical card. What would discharge it is unchanged: the cosmological constant as a residue is derived; the map from the record's steps to a physical clock is not.

Remark (The free seats)*.* That the free seats are the weakly interacting part of the construction: present in every instance, read by none, and accumulating once the local model is handed in. That the first thirteen steps of the middle card are the flat region of the sinc integral, which holds through the factor $1/13$, so that a search may take them as computed and begin at the fourteenth. That the middle card's least action, closing at degree three, is the bending energy of an elastic strip through its knots. *Proposed*, each: the counts agree, and no map between the two sides is derived.

Remark (Considered harmless)*.* The objection to the unrestricted jump [4] is that it severs the state of a process from the text of its program: after a jump, where the process is cannot be read from where the program is. $\mathbf{\Omega}$ is the jump to itself, and in it the two coincide at every step, because the step is the program. An unbranched loop asserts no value beyond its execution: a loop that wrote a second symbol would be a computation, and a loop that writes one symbol forever is a clock, which asserts only that another entry was written. The middle card's press check reads the head

of a closed record once and decides between two readings; it never evaluates a loop, which has no normal form and so no head to read. *Measured*, extracted from the declarations, for the press check; *proposed*, that the loop so read is a clock.

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